AMINE RHAMMI

MECHANICAL ENGINEERING GRADUATE CANDIDATE | ASSET INTEGRITY, RELIABILITY & PROJECT DELIVERY

+1 514 559 3364 | aminecana@hotmail.com | amine.rhammi@esso.ca

Ready to relocate to the UK under UK Youth Mobility Scheme visa | No sponsorship required

PROFESSIONAL PROFILE

B.Eng. Mechanical Engineering (Mechatronics) candidate graduating December 2027 combining 3 years of industrial experience through six placements across oil and gas, mining, pulp and paper, metallurgy, and manufacturing. Currently supporting mobile-equipment reliability at ExxonMobil Imperial Oil through maintenance readiness by combining SAP/CMMS evidence, RCM/FMECA/RCA, Weibull life-data analysis, condition monitoring, equipment strategy and practical field documentation. Earlier refinery and plant work covers pumps, valves, BOMs, MOC, P&IDs, commissioning, lubrication and CBM. Fluent in English/French and committed to progressing toward IMechE Chartership and a long-term UK engineering career.

EDUCATION

UNIVERSITÉ DU QUÉBEC À TROIS-RIVIÈRES (UQTR) | Trois-Rivières, Quebec, Canada — **Expected December 2027**

Bachelor of Engineering, Mechanical Engineering, Mechatronics concentration

ROSEMONT COLLEGE | Montreal, Quebec, Canada — **Completed**

Diploma of College Studies (DEC), Mathematics & Computer Science

CORE CAPABILITIES

Reliability & asset management: RCM; RCA/RCFA (TapRooT);FMECA; Weibull MLE; PM/ PdM optimisation; asset criticality; spares and lifecycle management; ISO 14224	**Maintenance systems & analytics: SAP PM; BOM/master data; Excel; Power Query; VBA; Power BI; Da ProcessBook; data historians**
Mechanical & field engineering: Pumps; compressors; heat exchangers; motors; conveyors; control valves/MOVs; lubrication; structural inspection; P&IDs; maintainability	**Projects, integrity & safety: MOC; commissioning; hazardous-service equipment; shutdown-planning;**

PROFESSIONAL ENGINEERING EXPERIENCE

IMPERIAL OIL | EXXONMOBIL |Calgary, Alberta, Canada — **May 2026 – Present**

Reliability & Maintenance Engineering, Internship

- Conducted an on-site structural survey of a newly introduced CAT 789 haul truck and developed a model-specific inspection reference defining standardized upper-chassis, lower-chassis, suspension, and dump-body inspection routes for reliability-engineering review.
- Collaborated with inspection teams to develop a structural assessment methodology for 81 heavy-haul truck frame assemblies, evaluating 2000+ inspection records against manufacturer support criteria to classify warranty activation eligibility, supporting a seven-figure cost-recovery case.
- Quantified associations between equipment loading exposure and structural-repair histories across an 83-truck fleet, presenting findings to reliability engineering to support risk-based inspection prioritisation and inform an ongoing maintenance-strategy review.
- Reconciled component-replacement histories across 150 trucks and 6 fleet models against planned maintenance intervals, providing planners and supply-chain with replacement-timing forecasts to support spare-parts positioning and PM strategy validation.
- Performed two-parameter Weibull MLE on 33 component populations across a 6-unit mining shovel fleet, classifying events as failed or right-censored, validating F/S accuracy against SAP work-order records, and estimating B10/B50 life parameters to distinguish age-related degradation from predominantly random failure behaviour.
- Established an ISO 14224-aligned equipment taxonomy across 199 mine-fleet assets (7 fleet types, 6.17M maintenance events database), providing reliability and maintenance-planning teams with standardised failure-mode and symptom coding that enabled consistent cross-fleet prioritization.

- Developed and deployed a jackknife-diagram classification framework across 228 fleet-system combinations and 7 fleet types, identifying top reliability drivers from downtime events to focus maintenance engineering effort where frequency × impact is highest.

VALERO Energy Corp. | Montreal, Quebec, Canada **May 2025 – Aug 2025**
Mechanical Engineering Intern, Reliability

- Developed the modification design of a hydrocarbon-drainage system incorporating two independent routes, isolation valves, vacuum-truck connections, and air admission to maintain a jet-fuel tank within its 300 Pa vacuum limit during controlled emptying.
- Prepared MOC documentation, updated P&IDs, and authored commissioning checklists for a submersible pump replacement in hazardous service, coordinating mechanical, structural, and electrical scope with the installation contractor and maintaining traceability for safe startup.
- Developed a low-pressure-drop magnetic pre-filtration concept for a 5,000 m^3/h hazardous-vapour system, targeting over 95% capture of ferrous particles above 10 µm at less than 1 inWC additional pressure drop.
- Reconciled field equipment information and OEM documentation against SAP PM bills of material across multiple pump families, identifying obsolete components, interchangeability opportunities, and master-data gaps affecting maintenance and spare-parts readiness.
- Performed field walkdowns across terminal loading stations to verify MOV and actuator nameplate data, including torque, speed, voltage, and mounting interface, building a terminal-wide register that resolved tag gaps and supported risk-based spare-parts planning.

KRUGER INC. | Trois-Rivières, Quebec, Canada **May 2024 – Aug 2024**
Mechanical Engineering Intern, Reliability

- Supported planning for the Paper Machine 4 shutdown by reviewing equipment condition with the reliability engineer and helping prioritise critical maintenance work for the outage.
- Prepared and helped conduct weekly condition-monitoring reviews for the pulp area and advanced Paper Machine, consolidating Spartakus vibration reports, identifying critical equipment, tracking repair actions, and communicating lubrication requirements to technicians.
- Audited lubrication and preventive-maintenance routes for duplicate tasks and coverage gaps, prioritised critical equipment, and developed a standardised Excel tool to track lubrication points, lubricant requirements, task frequencies, and completion status.
- Standardised more than 9,000 roller-history records to calculate operating duration from installation and removal dates and developed a belt-management tool incorporating tension calculations for plant fans and agitators.

RIO TINTO | Sorel-Tracy, Quebec, Canada **May 2023 – Aug 2023**
Mechanical Engineering Intern, Reliability

- Designed mechanical fixtures and protective guards in SolidWorks and AutoCAD, completed bending-stress, deflection, and weld-sizing calculations, and surveyed, designed, and field-verified process-piping elbow modifications to reduce pressure drop.
- Collected evidence for a fire-incident investigation caused by a water leak, developing and implementing mitigation measures to prevent recurrence.
- Developed conveyor maintenance plans and updated AssetWise and SAP PM parts records to improve equipment traceability and maintenance readiness.

SOUCY | Drummondville, Quebec, Canada **May 2022 – Aug 2022**
Mechanical Engineering Intern, Maintenance

- Coordinated the annual inspection of overhead cranes and lifting equipment, preparing the asset scope, accompanying the external inspector, reviewing reports for completeness, and tracking identified deficiencies through corrective maintenance and documented closure.
- Developed an oil-waste handling and storage system with the site's environmental-services provider, eliminating unsafe outdoor accumulation of open containers and reducing handling costs by more than CAD 30,000$.
- Supported implementation of a new preventive-maintenance system.
- Designed, fabricated, and installed machine-safety guards with internal stakeholders and supported department-efficiency projects.
- Supported electromechanical technicians in troubleshooting faults on rubber-track production equipment, coordinating replacement-part requirements with suppliers and purchasing to support timely maintenance execution.

RIO TINTO | Sorel-Tracy, Quebec, Canada **May 2021 – Apr 2022**
Mechanical Engineering Intern, Process Optimisation

- Developed equipment-monitoring interfaces for the upgraded-slag plant to support reliability initiatives and predictive-maintenance work.
- Assisted reliability engineers with RCA and maintenance-plan development and conducted compliance reviews to identify improvement and investment opportunities.

- Contributed to FMECA for the acid-regeneration plant, improving visibility of equipment failure modes, effects, and operational criticality.

SELECTED EXXONMOBIL RELIABILITY & ENGINEERING TRAINING

TapRooT Root Cause Analysis; Probability & Statistics for Reliability; RCM Equipment Strategy Development; Fixed Equipment Strategy Development; Risk-Based Work Selection Analysis; Pump, Compressor, Heat Exchanger and Control Valve Troubleshooting.

ADDITIONAL INFORMATION

Languages: Fluent English and French

Professional development: Committed to an IMechE Chartership pathway

Mobility and work authorisation: Prepared to relocate to the Southampton/Fawley area for the September 2027 start; no sponsorship required at commencement;